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Profile

I build systems that operate under uncertainty across finance, healthcare, legal reasoning, and physical simulation. My work combines machine learning, backend engineering, and physics-based modeling to create systems that are measurable, explainable, and grounded in real-world constraints.

High Impact Systems

LexProbe — Citation-Verified Legal AI

RAG-based legal AI system with a citation audit layer.

Pipeline: Query → Retrieval → Context → LLM → Audit → Verified Output

Work: Citation normalization, token overlap verification, removal of hallucinated citations.

Insight: Trust in legal AI requires verifiable outputs.

GeoQuant — Quantitative Trading System

End-to-end ML system for financial prediction and trading strategy evaluation.

Pipeline: Data → Feature Engineering → Model → Signals → Backtesting → Metrics

Work: Feature engineering (technical + news), ML prediction system, backtesting integration with cost and slippage.

Insight: Profitability depends on execution realism, not just prediction accuracy.

Health AI — Explainable Risk System

Health risk prediction system with explainability and safety layers.

Pipeline: Input → Model → Explanation → Confidence → Output

Work: Risk scoring system, ML integration, explainability layer, safety constraints.

Insight: Explainability and safety are critical in healthcare systems.

Simulation & Applied Physics

Propeller Performance Simulation

Physics-based simulation analyzing thrust, RPM, pitch, and efficiency.

Outputs: Graphs, CSV data, comparative analysis.

Vortex Ring State Simulation

Simulation of aerodynamic instability in drones.

Work: Modeled airflow behavior, analyzed instability conditions, validated through research.

Outputs: Graphs, logs, comparative simulations.

Engineering & AI Projects

Computer Vision: Gesture control, sign language recognition.

DNS-based Network Ad Blocker.

Voice Assistant (Jarvis).

Autonomous Systems & Hardware

Autonomous drones (quad/hex),

plant watering system

rovers

home automation.

Raspberry Pi NAS and distributed computing experiments.

Research Work

Blade Momentum Element Theory: pitch, RPM, blade count.

Vortex ring formation research.

Internships

Indian Institute of Science (IISc): Vortex ring research.

Vegam Co: Air filtration system, RAG pipeline (NL to SQL).

Entrepreneurship

Elevyx: Backend systems.

Athera: Workflow automation platform.

Orqis: (co-Founder) end to end pipeline fix for all IDEs through detection and auto-prompting fixes and github prs

Sports

Competitive skating, football (CBSE tournaments, Goa Globe), badminton, swimming.

Model United Nations

1 year experience in research, debate, diplomacy.

Social Work

50+ hours of community service.